External links

History

Glucose was first isolated from raisins in 1747 by the German chemist Andreas Marggraf.[8][9] Glucose was discovered in grapes by Johann Tobias Lowitz in 1792 and recognized as different from cane sugar (sucrose). Glucose is the term coined by Jean Baptiste Dumas in 1838, which has prevailed in the chemical literature. Friedrich August Kekulé proposed the term dextrose (from Latin dexter = right), because in aqueous solution of glucose, the plane of linearly polarized light is turned to the right. In contrast, D-fructose (a ketohexose) and L-glucose turn linearly polarized light to the left. The earlier notation according to the rotation of the plane of linearly polarized light (*d* and *l*-nomenclature) was later abandoned in favor of the D- and L-notation, which refers to the absolute configuration of the asymmetric center farthest from the carbonyl group, and in concordance with the configuration of D- or L-glyceraldehyde.[10][11]

Since glucose is a basic necessity of many organisms, a correct understanding of its chemical makeup and structure contributed greatly to a general advancement in organic chemistry. This understanding occurred largely as a result of the investigations of Emil Fischer, a German chemist who received the 1902 Nobel Prize in Chemistry for his findings.[12] The synthesis of glucose established the structure of organic material and consequently formed the first definitive validation of Jacobus Henricus van 't Hoff's theories of chemical kinetics and the arrangements of chemical bonds in carbon-bearing molecules.[13] Between 1891 and 1894, Fischer established the stereochemical configuration of all the known sugars and correctly predicted the possible isomers, applying van 't Hoff's theory of asymmetrical carbon atoms. The names initially referred to the natural substances. Their enantiomers were given the same name with the introduction of systematic nomenclatures, taking into account absolute stereochemistry (e.g. Fischer nomenclature, D/L nomenclature).

For the discovery of the metabolism of glucose Otto Meyerhof received the Nobel Prize in Physiology or Medicine in 1922.[14] Hans von Euler-Chelpin was awarded the Nobel Prize in Chemistry along with Arthur Harden in 1929 for their "research on the fermentation of sugar and their share of enzymes in this process".[15][16] In 1947, Bernardo Houssay (for his discovery of the role of the pituitary gland in the metabolism of glucose and the derived carbohydrates) as well as Carl and Gerty Cori (for their discovery of the conversion of glycogen from glucose) received the Nobel Prize in Physiology or Medicine.[17][18][19] In 1970, Luis Leloir was awarded the Nobel Prize in Chemistry for the discovery of glucose-derived sugar nucleotides in the biosynthesis of carbohydrates.[20]

Chemical properties

With six carbon atoms, it is classed as a hexose, a subcategory of the monosaccharides. D-Glucose is one of the sixteen aldohexose stereoisomers. The D-isomer, D-glucose, also known as dextrose, occurs widely in nature, but the L-isomer, L-glucose, does not. Glucose can be obtained by hydrolysis of carbohydrates such as milk sugar (lactose), cane sugar (sucrose), maltose, cellulose, glycogen, etc. It is commonly commercially manufactured from cornstarch by hydrolysis via pressurized steaming at controlled pH in a jet followed by further enzymatic depolymerization.[21] Unbonded glucose is one of the main ingredients of honey. All forms of glucose are colorless and easily soluble in water, acetic acid, and several other solvents. They are only sparingly soluble in methanol and ethanol.

Structure and nomenclature

Glucose is a monosaccharide with formula $C_6H_{12}O_6$ or H-(C=O)-$(CHOH)_5$-H, whose five hydroxyl (OH) groups are arranged in a specific way along its six-carbon back. Glucose is usually present in solid form as a monohydrate with a closed pyran ring (dextrose hydrate). In aqueous solution, on the other hand, it is an open-chain to a small extent and is present predominantly as α- or β-pyranose, which partially mutually merge by mutarotation. From aqueous solutions, the three known forms can be crystallized: α-glucopyranose, β-glucopyranose and β-glucopyranose hydrate.[22] Glucose is a building block of the disaccharides lactose and sucrose (cane or beet sugar), of oligosaccharides such as raffinose and of polysaccharides such as starch and amylopectin, glycogen or cellulose. The glass transition temperature of glucose is 31 °C and the Gordon-Taylor constant (an experimentally determined constant for the prediction of the glass transition temperature for different mass fractions of a mixture of two substances)[23] is 4.5.[24]

	5BC%40H%5D%28%5BC%40%40H%5D%28%5BC%40H%5D%28%5BC%40H%5D%28O1%29O%29O%29O%29O%29O)
3DMet	B01203 (http://www.3dmet.dna.affrc.go.jp/cgi/show_data.php?acc=B01203)
Abbreviations	Glc
Beilstein Reference	1281604
ChEBI	CHEBI:4167 (https://www.ebi.ac.uk/chebi/searchId.do?chebiId=4167) ✓
ChEMBL	ChEMBL1222250 (https://www.ebi.ac.uk/chembldb/index.php/compound/inspect/ChEMBL1222250) ✓
ChemSpider	5589 (http://www.chemspider.com/Chemical-Structure.5589.html) ✓
EC Number	200-075-1
Gmelin Reference	83256
IUPHAR/BPS	4536 (http://www.guidetopharmacology.org/GRAC/LigandDisplayForward?tab=summary&ligandId=4536)
KEGG	C00031 (https://www.kegg.jp/entry/C00031) ✓
MeSH	Glucose (https://www.nlm.nih.gov/cgi/mesh/2014/MB_cgi?mode=&term=Glucose)
PubChem CID	5793 (https://pubchem.ncbi.nlm.nih.gov/compound/5793)
RTECS number	LZ6600000
UNII	5SL0G7R0OK (https://fdasis.nlm.nih.gov/srs/srsdirect.jsp?regno=5SL0G7R0OK) ✓
InChI	
SMILES	
Properties	
Chemical	$C_6H_{12}O_6$